Who Builds What: Every Feature Explained

Campus Plus: a campus request board where students post what they need and sellers compete with lower prices (reverse bidding). 21-day plan.

How to read this: each member has a list of features. **"What it means"** explains the feature in plain words, **"Example"** shows how it looks, and **"When"** is the project day range from the tracker. Features marked **(optional)** are cut first if we run late.

Member 1: Database foundation and documents

Tables you own: categories, departments, pickup_locations
Works closely with: everyone (all tables connect to the shared design)

Feature	What it means	Example	When
ER merge	Combine everyone's draft tables into one diagram and check that names and links match.	The offers table links to requests with request_id.	Days 2-3
Schema runner (run_all.sql)	One file that builds the whole database in the right order.	One command creates all tables.	Day 4
Migrations	After the ER freeze, every change goes into a new numbered file. Old files are never edited.	002_add_rating_comment.sql	When needed
Master data	Fixed lists used everywhere: departments, categories with sub-categories, pickup points.	Electronics, then Arduino. Pickup: library, hostel gate.	Days 4-5
SQL reports	Queries that turn data into numbers for the admin and the demo.	Average savings, most requested categories, top-rated sellers, requests with no offers.	Days 12-15
Report pages	Admin pages that show those reports.	A table of top sellers.	Days 15-17
Repo and shared files	Set up GitHub, folders and README. Only you edit the shared files: server.js, db.js, main.css, package.json.	A teammate needs a new package, so you add it.	All days
Final data and docs	Merge everyone's sample data. Prepare the final ER, API list and report screenshots.	Demo data that looks real.	Days 19-20

Member 2: People and trust

Tables you own: users, ratings, notifications
Works closely with: Member 3 (request pages), Member 4 (offer pages), Member 5 (deals)

Feature	What it means	Example	When
Users	The table and code that store each person's details.	Name, college email, department, role.	Day 4
Register and login	Sign up and sign in with a college email. Passwords are stored safely (hashed).	Only verified students can use the site.	Days 5-6
Roles	Different permissions for different users.	Student, senior, admin. Only admin sees moderation pages.	Day 5
Login and register pages	The screens for signing up and logging in.	A form with email and password.	Day 6
Ratings	After a deal, the buyer rates the seller.	4 out of 5 stars.	Days 14-15
Rating on offers	Show each seller's average rating next to their offer.	"Seller rating: 4.5" beside an offer.	Day 15
In-app notifications (optional)	Alerts inside the site, like a bell icon. No email or SMS.	"A new offer on your request."	Day 16
Profile page (optional)	A page with the user's own activity.	My requests, my offers, my rating.	Day 17

Member 3: Requests and browsing

Tables you own: requests, seller_items
Works closely with: Member 1 (categories), Member 4 (offers shown on the request page)

Feature	What it means	Example	When
Requests	A buyer posts what they need, with a budget and a deadline.	"Need a scientific calculator, max Rs. 600, by Friday."	Days 4-6
Request pages	The pages that list requests and show one request in detail.	A list of open requests, and a details page.	Days 6-10
Seller items	Sellers list items they already have. This also helps when few sellers answer a request.	"Arduino Uno kit, Rs. 800."	Days 8-9
Search and filter	Find requests quickly.	Filter by category, budget range or deadline.	Days 13-14
Polish your pages	Check every page for broken links and clear error messages.	A message if the form is incomplete.	Day 17

Member 4: The bidding engine (the heart of the project)

Tables you own: offers, offer_history
Works closely with: Member 3 (request page), Member 5 (transaction after a winner is chosen)

Feature	What it means	Example	When
Place an offer	A seller offers a price on a request.	"I can give it for Rs. 450."	Day 8
Offer checks	Rules that reject invalid offers with a clear message.	Must be lower than the current lowest, within budget, not on your own request, and the request must be open.	Day 8
Offer history	Every offer and every price drop is saved.	Rs. 550, then 500, then 450, with times.	Day 8
Show offers	Show the offers and price history on the request page.	Lowest offer on top, history below.	Day 10
Closing	When the deadline passes, the request closes by itself.	After Friday 6 PM, no new offers.	Day 9
Choosing the winner	The buyer picks the best offer. It is not always the cheapest, since condition and pickup also matter.	The chosen offer is saved and triggers the transaction.	Day 10
Postman tests	Test the offer APIs before connecting pages.	A saved test collection.	Day 11

Member 5: Deals, safety and testing

Tables you own: transactions, handover_records, complaints, penalties
Works closely with: Member 4 (winner chosen), Member 2 (ratings and users)

Feature	What it means	Example	When
Share the tracker	Set up the Google Sheet and give everyone Editor access.	All five members can edit it.	Day 1
Transactions	A record of the agreed deal with a simulated payment status. No real payments.	Buyer, seller, amount, status: pending, paid or failed.	Day 10
Handover	Tracking the physical exchange of the item.	Status moves: agreed, handed over, confirmed.	Day 13
Complaints	A user can report a problem.	"The seller did not show up."	Day 15
Penalties (optional)	Admin actions against repeat offenders.	A warning or a temporary block from making offers.	Day 16
Admin pages	Screens where the admin reviews complaints and moderates.	A table of open complaints with Approve and Reject buttons.	Day 17
Test checklist	A list of things to check at the end. The whole team runs it.	"Can a seller offer higher than the lowest? It should fail."	Day 18
Demo script	The story and sample data for the final presentation.	Post a request, sellers compete, buyer picks, deal completes.	Day 19

How the pieces connect

M2 user logs in → **M3** posts a request → **M4** sellers make lower offers, request closes, buyer chooses → **M5** transaction and handover → **M2** buyer rates seller → **M1** reports show the results from the merged database.

Remember: build only your own features and files. If you need something from another member's part, ask them or post in the group. Update your Status in the Google Sheet every day. Total: about 14 tables. Member 4 has the hardest logic, Member 5 has the most separate parts, and Member 1 has the lightest load but owns the shared pieces.